



FUNDAMENTALS IN RFID

Part 2

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Good news

Heckling is encouraged

Warning

New allocation of sessions

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New plan



TOPIC	SESSION
Answers to homework	1
RFID protocols	2,3
Dense reader problems and issues	4
EPCglobal network elements and functions	4
Security and authentication	5
Adelaide Auto-ID research program	6,7
Conclusions	7

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PART 9

ANSWERS TO HOMEWORK

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Outline



◆ Highlights only given here

◆ See separately distributed notes for detail

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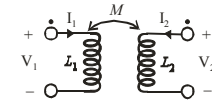
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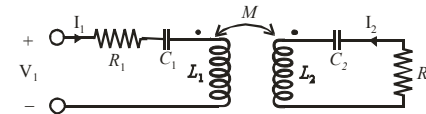


Question D8:1





$$k = \frac{M}{\sqrt{L_1 L_2}} \quad \begin{aligned} V_1 &= j\omega L_1 I_1 + j\omega M I_2 \\ V_2 &= j\omega M I_1 + j\omega L_2 I_2 \end{aligned}$$



$$P_2/P_1 = k^2 Q_1 Q_2$$

The weak coupling approximation is made when we assume that the voltage induced in the primary by the secondary current is small in relation to V_1 . With this understanding, and a knowledge of definitions of quality factor and the fact that at resonance the impedances of the capacitors cancel those of the inductors, and a knowledge of how to calculate the power dissipated by a sinusoidal current in a resistor, only a few lines of algebra will produce the power transfer result above.

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Question D8:2



$$V_c = \frac{[\text{Reactive power flowing in the untuned label coil when it is short circuited}]}{[\text{Volume density of reactive power created by the interrogator at the label position}]}$$

$$V_d = \frac{[\text{Reactive power flowing in the inductor of the interrogator field creation coil}]}{[\text{Volume density of reactive power created by the interrogator at the label position}]}$$

The re-expressed power transfer relation is below. It is easily produced from the previous relation when we realize (a) that the real power P_2 dissipated in R_2 is Q_2 times the reactive power flowing in the inductor of the tag when it is assumed to be lossless, un-tuned and short-circuited as is assumed in the definition of V_c , and (b) that a similar statement can be made about P_1 , and the denominators in the definitions of V_c and V_d are the same.

$$\frac{P_2}{P_1} = \frac{V_c}{V_d} Q_1 Q_2 \quad \frac{P_2}{P_1} = 1.075 \times 10^{-3}$$

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Question D9



The EIRP in the direction of maximum gain = $5 \times 1.5 \text{ mW}$

The power density S_r per unit area at 10 m = $5 \times 1.5 \text{ mW} \times \frac{1}{4\pi 10^2} = 6 \mu\text{Wm}^{-2}$

$$H_\theta = \frac{\beta^2 j \omega \mu_0 M}{4\pi \eta} \left(\frac{j}{(\beta r)} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3} \right) e^{-j\beta r} \sin \theta$$

Radial component of Poynting vector $S_r = \frac{\eta |H|^2}{2}$ in Wm^{-2}

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Question D9 continued



Radial component of Poynting vector $S_r = \frac{\eta |H|^2}{2}$ in Wm^{-2}

$$S_r = \frac{\eta |H|^2}{2} \text{ in } \text{Wm}^{-2}$$

$$\frac{2S_r}{\eta} = |H|^2$$

$$H = \sqrt{\frac{2S_r}{\eta}}$$

$$H = 178 \mu\text{Am}^{-1}$$

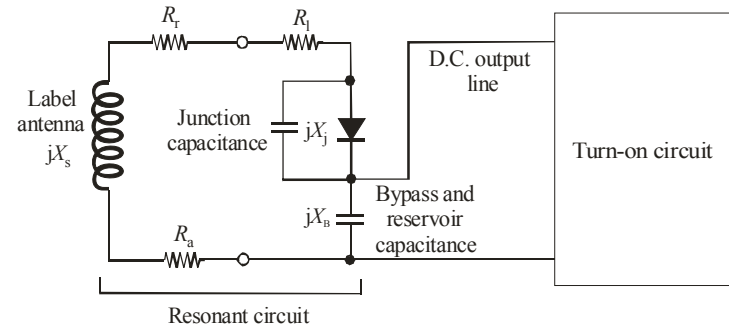
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Question D10



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Question D10 continued



$$P_r = \frac{2\pi f C |V_{r.m.s.}|^2}{Q}$$

$$\text{Power flow per unit area} = \frac{P_r}{A_{er}} = \frac{4\pi P_r}{g_r \lambda^2}$$

$$\text{Power flow per unit area} = \frac{g_t P_t}{4\pi r^2}$$

Using $g_t P_t$ of 4 W we obtain $r = 5.218 \text{ m}$

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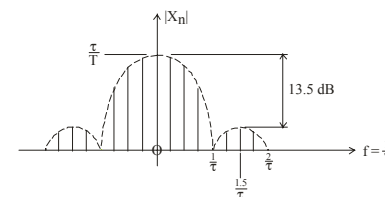


Question D11



$$\tau = 10\mu\text{s}$$

$$T = 80\mu\text{s}$$



$$X_n = \left(\frac{\tau}{T}\right) \frac{\sin(\pi f_n \tau)}{\pi f_n \tau}$$

Note the amplitude is also reduced by the dip depth of 20%

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Question D11 continued



The relative amplitude of the central peak is composed of the duty cycle of 1/8, (-18 dB) and the 20% dip depth (-14dB), a total of -32 dB. The first spectral line out of the central peak of the allowed spectrum is at 12.5 kHz, for which the $\sin x/x$ function has the value $[\sin(\pi/8)]/(\pi/8)$ i.e. -1.1 dB, changing the total to -33.1 dB.

The regulations are shown overleaf. They show that in Europe this form of signalling complies. In the USA it almost complies under the analysis conducted here, and would almost certainly comply if additional band limiting that is almost certainly present in the system were taken into account.

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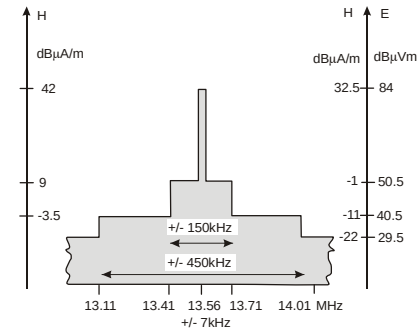


Question D11 continued



EUROPEAN REGULATIONS
FIELD STRENGTH
@ D = 10 m

REVISED FCC PETITION
FIELD STRENGTH
@ D = 30 m



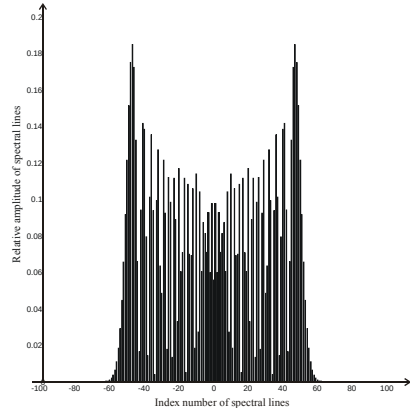
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Question D11 continued



Spectral spreading with frequency modulation

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Question D11 continued



Examination of the results Appendix C shows that after a frequency modulation of the carrier by the parameters given therein is applied, the largest spectral line has been reduced to a relative amplitude of 0.185. This is a 14.6 dB reduction in power level. It is reasonable to assume that all of the signalling sidebands produced by the amplitude modulation are reduced by the same amount.

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PART 10

RFID PROTOCOLS

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What is a protocol?

- ◆ Signalling waveforms
- ◆ Command set
- ◆ Operating procedure
- ◆ A back end interface

whereby the identities of a population of tags in the field of a reader may be determined, and the population otherwise managed.

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Constraints on protocols



- ◆ Electromagnetic compatibility regulations
 - Differ with frequency range and jurisdiction
 - Some convergence is occurring
- ◆ Reader to reader interference
 - Readers confusing tags
 - Readers blocking other reader receivers
- ◆ Simplicity (as reflected in chip size)
 - Maybe that influences reliability as well

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Auto-ID Center protocols

- ◆ The Auto-ID Center defined
 - The Class 1 UHF protocol
 - The Class 1 HF protocol
 - The Class 0 UHF protocol
- ◆ EPCglobal has defined in addition
 - Generation 2 UHF protocol

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Why are they different?



- ◆ Different field properties at HF and UHF
 - Near and far field – different field confinement
 - Different field penetration in materials
 - Different silicon circuit possibilities and costs
 - Different electromagnetic regulations
- ◆ Read only memory technologies enable miniaturisation
- ◆ A high performance UHF system was available and was modified by the Center to manage privacy concerns

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Protocols: the major divide



- ◆ Tree walking
 - More forward link signalling
 - Prolonged periods of interrupted signalling
 - Partial information of tag population remains relevant
- ◆ Adaptive round (terminating aloha)
 - Less forward link signalling
 - Long periods of un-modulated reader carrier
 - Reader signalling is less
 - No information from one response about others

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Characteristics: similarities



- ◆ Both can select subsets of tags for participation
- ◆ Overt selection may reveal to an eavesdropper what is selected
- ◆ Forms of less overt selection are possible
- ◆ Tag “sleeping” has a role in both

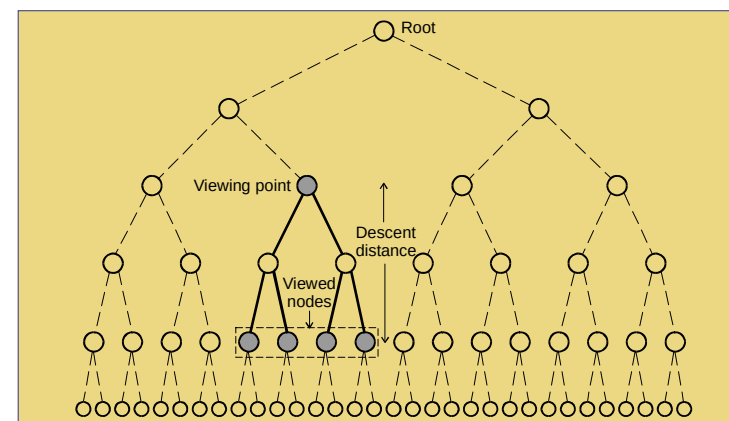
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Tree scanning concepts



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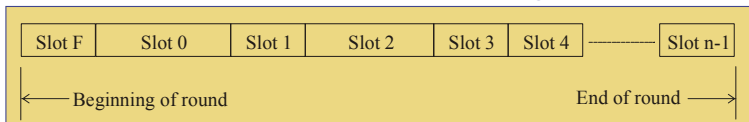
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Adaptive round concepts



- ◆ Labels reply once per *round*, in randomly chosen slots
- ◆ A group of n slots forms a *round*
- ◆ The number of slots in a round varies as needed
- ◆ Tags with already collected replies are moved to slot F, or silenced in some way.



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Adaptive round concepts: 2



- ◆ What we saw was the Class 1 Generation 1 HF protocol
- ◆ Adaptive round concept now appears in many places
- ◆ Next protocol is a complex example of an adaptive round

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The C1G2 protocol



- ◆ Labels have an adjustable probability of replying on each query or repeated query
- ◆ Probability is adjusted to about a third
- ◆ Empty slots, singly occupied slots and multiply occupied slots are roughly equi-probable
- ◆ A wide range of forward and reverse signalling parameters are defined
- ◆ Some of them allow for narrow band reply signalling well separated from the interrogation carrier

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C1G2 Features

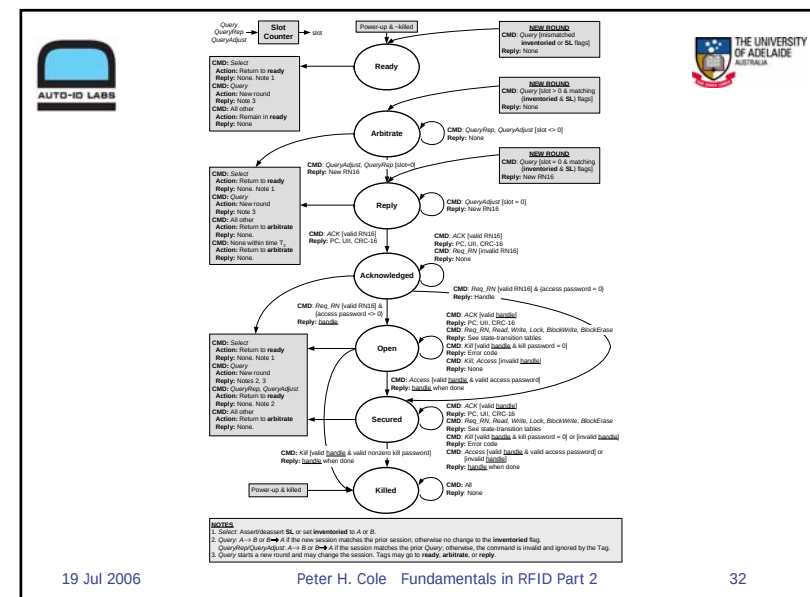
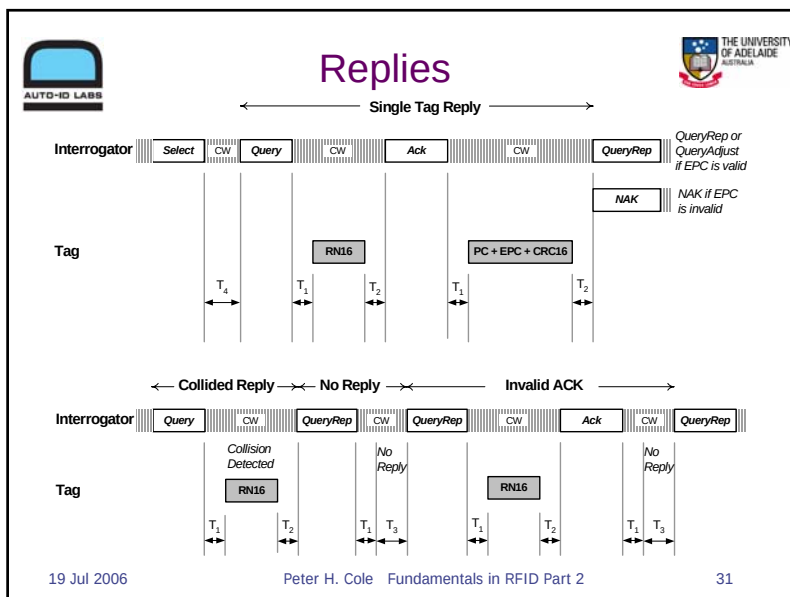
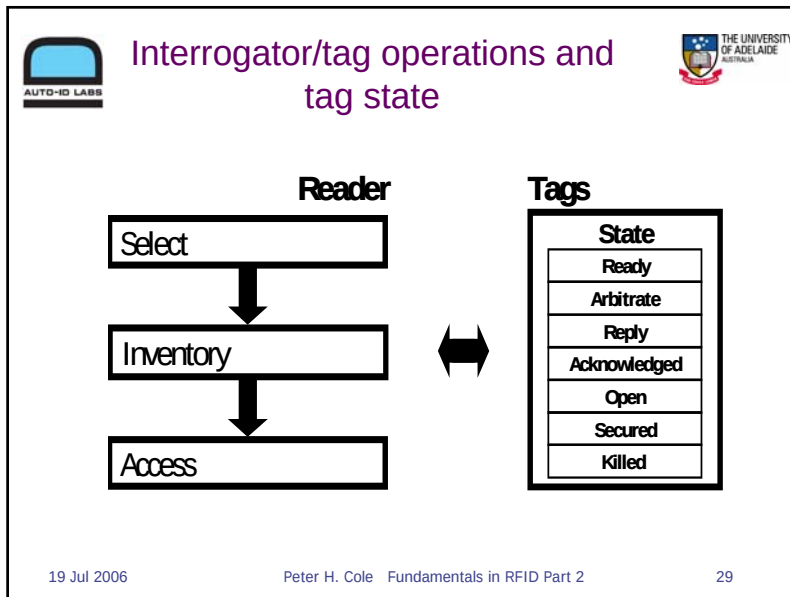


- ✦ Tag must be able to communicate from 860 MHz to 960 MHz
- ✦ Tags must understand 3 different modulation schemes
 - Double Sideband Amplitude Shift Keying DSB-ASK
 - Single Sideband Amplitude Shift Keying SSB-ASK
 - Phase Reversal Amplitude Shift Keying PR-ASK
- ✦ Coding is by Pulse Interval Encoding (PIE)
- ✦ T=>R data rates 40, 80, 160, 320 and 640 kbits
- ✦ Selection
- ✦ Access & Kill Passwords
- ✦ EPC up to 256 bits
- ✦ Dense reader channelised signalling

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C1G2 details

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R=>T Signalling

- ◆ An Interrogator communicates with one or more Tags by modulating an RF carrier using DSB-ASK, SSB-ASK, or PR-ASK with PIE encoding.
- ◆ Interrogators shall use a fixed modulation format and data rate for the duration of an inventory round.
- ◆ The Interrogator sets the data rate by means of the preamble that initiates the round.

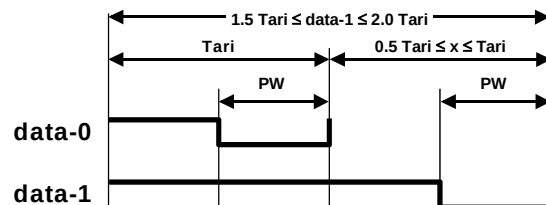
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PIE symbols

- ◆ Tari is the reference time interval for interrogator-to-tag signalling, and is the duration of a data-0.
- High values represent transmitted CW;
- Low values represent attenuated CW.
- Tari varies between 6.25μs and 25μs, inclusive, but fixed for a round.



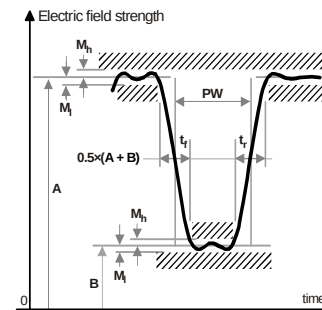
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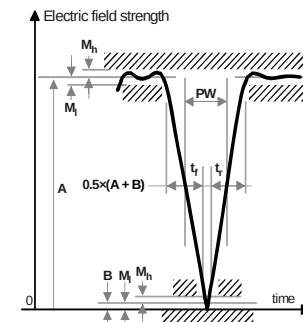
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R=>T RF envelope

ASK Modulation



PR-ASK Modulation



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R=>T Signalling



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R=>T preamble and frame-sync



- ◆ An interrogator shall begin all R=>T signalling with either a preamble or a frame-sync
 - A preamble shall precede a *Query* command and denotes the start of an inventory round.
 - All other signalling shall begin with a frame-sync.

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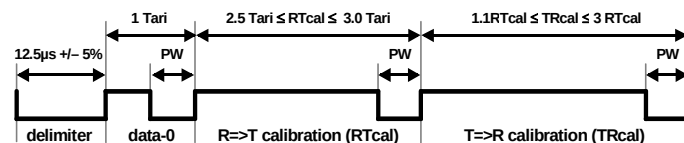
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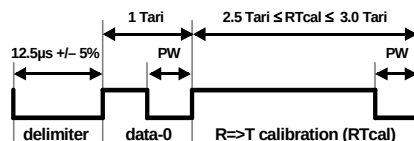
R=>T preamble and frame-sync



R=>T Preamble



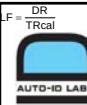
R=>T Frame-Sync



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Preamble

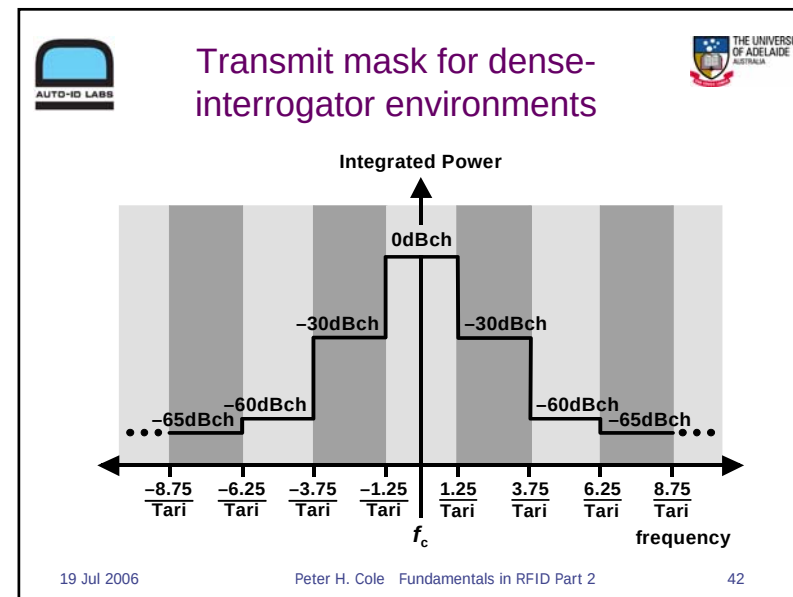
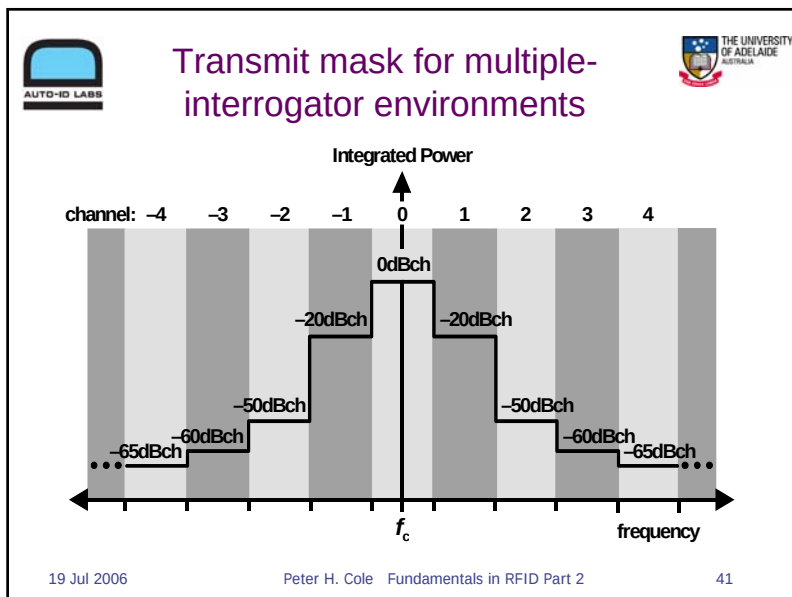


- ◆ **RTcal:**
 - RTcal equals the length of a data-0 symbol plus the length of a data-1 symbol.
 - Tags measure the length of RTcal and compute $pivot = RTcal / 2$.
 - Tags interpret subsequent interrogator symbols shorter than *pivot* to be data-0, and subsequent interrogator symbols longer than *pivot* to be data-1.
- ◆ **TRcal:**
 - The interrogator specifies a tag's backscatter link frequency (its FM0 datarate or the frequency of its Miller subcarrier) using the TRcal and divide ratio (DR) in the preamble and payload
 - Link Frequency = $DR/TRcal$

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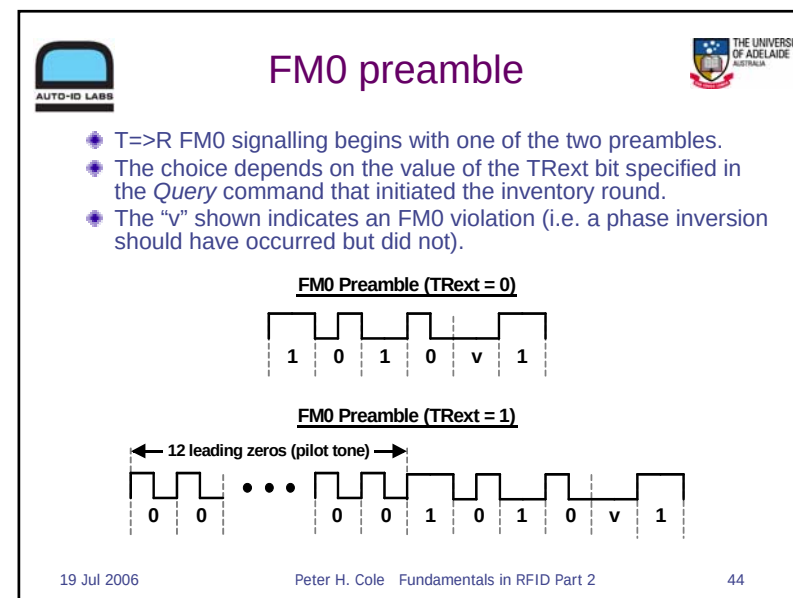
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Tag-to-interrogator (T=>R) communications

- ◆ Data modulation
 - Tag backscatter shall use ASK and/or PSK modulation (up to tag vendor), but Interrogators are capable of demodulating either type.
- ◆ Data encoding
 - Tags shall encode the backscattered data as either FM0 baseband or Miller modulation of a subcarrier at the data rate. The interrogator commands the encoding choice.

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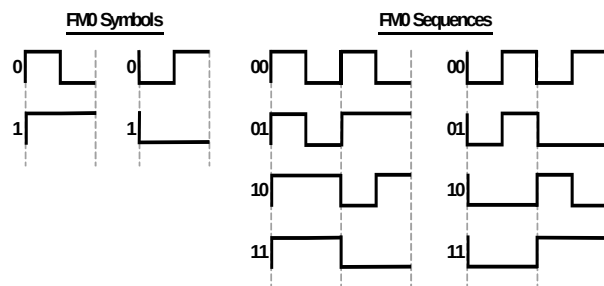




FM0



- FM0 (bi-phase space) encoding.
 - FM0 inverts the baseband phase at every symbol boundary; a data-0 has an additional mid-symbol phase inversion.



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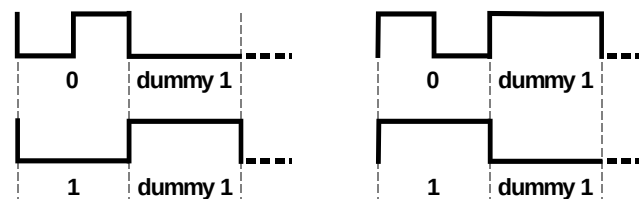


FM0 Termination



FM0 signalling always ends with a “dummy” data-1 bit

FM0 End-of-Signaling



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Miller encoding



- Baseband Miller inverts its phase between two data-0s in sequence.
- Baseband Miller also places a phase inversion in the middle of a data-1 symbol.
- The Miller sequence shall contain exactly two, four, or eight subcarrier cycles per bit, depending on the M value specified in the *Query* command that initiated the inventory round.
- T=>R subcarrier signalling shall begin with one of the two preambles. The choice depends on the value of the T_{Text} bit specified in the *Query* command that initiated the inventory round.

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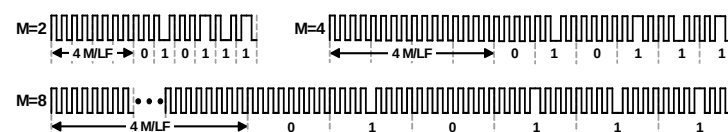
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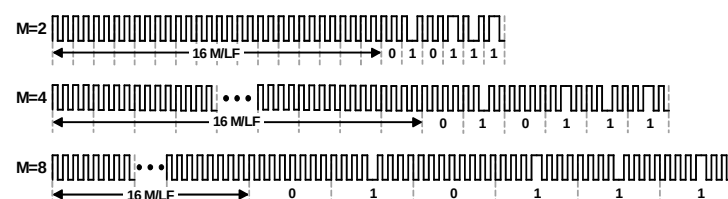
Preamble



Miller Preamble (T_{Text} = 0)



Miller Preamble (T_{Text} = 1)



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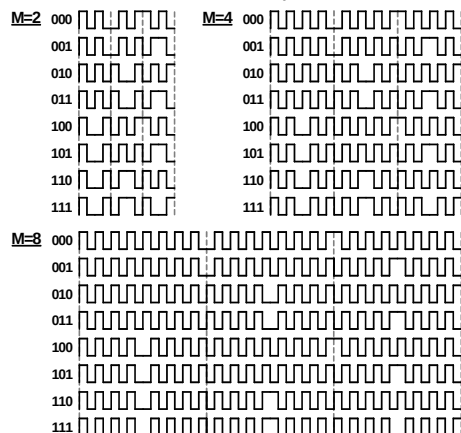
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Miller Subcarrier Sequences



Miller Subcarrier Sequences



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Data Rates



DR: Divide Ratio	TR_{cal}^a ($\mu s \pm 1\%$)	LF: Link Frequency (kHz)	Frequency Tolerance FT (nominal temp)	Frequency Tolerance FT (extended temp)	Frequency variation during backscatter
64/3	33.3	640	$\pm 15\%$	$\pm 15\%$	$\pm 2.5\%$
	$33.3 < TR_{cal} < 66.7$	$320 < LF < 640$	$\pm 22\%$	$\pm 22\%$	$\pm 2.5\%$
	66.7	320	$\pm 10\%$	$\pm 15\%$	$\pm 2.5\%$
	$66.7 < TR_{cal} < 83.3$	$256 < LF < 320$	$\pm 12\%$	$\pm 15\%$	$\pm 2.5\%$
	83.3	256	$\pm 10\%$	$\pm 10\%$	$\pm 2.5\%$
	$83.3 < TR_{cal} < 133.3$	$160 < LF < 256$	$\pm 10\%$	$\pm 12\%$	$\pm 2.5\%$
	$133.3 < TR_{cal} < 200$	$107 < LF < 160$	$\pm 7\%$	$\pm 7\%$	$\pm 2.5\%$
8	$200 < TR_{cal} < 225$	$95 < LF < 107$	$\pm 5\%$	$\pm 5\%$	$\pm 2.5\%$
	$17.2 \leq TR_{cal} < 25$	$320 < LF \leq 465$	$\pm 19\%$	$\pm 19\%$	$\pm 2.5\%$
	25	320	$\pm 10\%$	$\pm 15\%$	$\pm 2.5\%$
	$25 < TR_{cal} < 31.25$	$256 < LF < 320$	$\pm 12\%$	$\pm 15\%$	$\pm 2.5\%$
	31.25	256	$\pm 10\%$	$\pm 10\%$	$\pm 2.5\%$
	$31.25 < TR_{cal} < 50$	$160 < LF < 256$	$\pm 10\%$	$\pm 10\%$	$\pm 2.5\%$
	50	160	$\pm 7\%$	$\pm 7\%$	$\pm 2.5\%$
	$50 < TR_{cal} \leq 75$	$107 \leq LF < 160$	$\pm 7\%$	$\pm 7\%$	$\pm 2.5\%$
	$75 < TR_{cal} \leq 200$	$40 \leq LF < 107$	$\pm 4\%$	$\pm 4\%$	$\pm 2.5\%$

^a Allowing two different TR_{cal} values (with two different DR values) to specify the same LF offers flexibility in specifying T_{ari} and R_{Tcal} .

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Tag-to-interrogator data rates



M: Number of subcarrier cycles per symbol	Modulation type	Data rate (kbps)
1	FM0 baseband	LF
2	Miller subcarrier	LF/2
4	Miller subcarrier	LF/4
8	Miller subcarrier	LF/8

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Three basic Operations

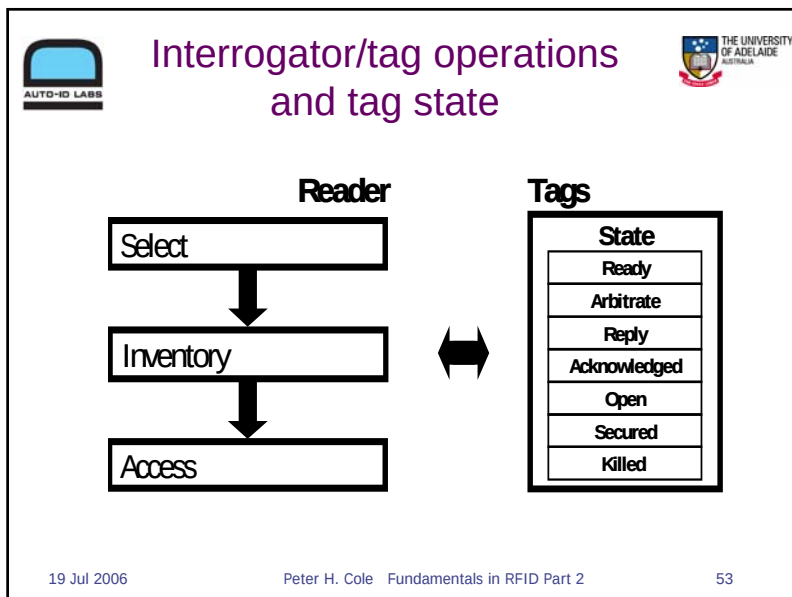


- ✦ **Select.**
 - The operation of choosing a tag population for inventory and access.
 - A *Select* command may be applied successively to select a particular tag population based on user-specified criteria.
- ✦ **Inventory.**
 - The operation of identifying tags.
 - An interrogator begins an inventory round by transmitting a *Query* command in one of four sessions. One or more tags may reply.
 - The interrogator detects a single tag reply and requests the PC, UII, and CRC-16 from the tag.
 - Inventory comprises multiple commands.
 - An inventory round operates in one and only one session at a time.
- ✦ **Access.**
 - The operation of communicating with (reading from and/or writing to) a tag.
 - An individual tag must be uniquely identified prior to access.
 - Access comprises multiple commands, some of which employ one-time-pad based cover-coding of the R=>T link.

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Selecting: 1

- The selection process employs a single command, *Select*, which an interrogator may apply successively to select a particular tag population based on user-defined criteria, enabling union ($\cup$), intersection ($\cap$), and negation ($\sim$) based tag partitioning.
- Interrogators perform $\cap$ and $\cup$ operations by issuing successive *Select* commands. *Select* can assert or deassert a tag's **SL** flag, or it can set a tag's **inventoried** flag to either *A* or *B* in any one of the four sessions.

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Selecting: 2

- Select* contains the parameters Target, Action, MemBank, Pointer, Length, Mask, and Truncate.
 - Target and Action indicate whether and how a *Select* modifies a tag's **SL** or **inventoried** flag, and in the case of the **inventoried** flag, for which session.
 - MemBank specifies if the mask applies to Ull, TID, or User memory. *Select* commands apply to a single memory bank. Successive *Selects* may apply to different memory banks.

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Selecting: 3

- Pointer, Length, and Mask: Pointer and Length describe a memory range. Mask, which must be Length bits long, contains a bit string that a tag compares against the specified memory range.
 - Truncate specifies whether a tag backscatters its entire Ull, or only that portion of the Ull immediately following Mask. Truncated replies are always followed by the CRC-16 in Ull memory 00h to 0Fh; a tag does not re-compute this CRC for a truncated reply.

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Selecting: 4



- ◆ By issuing multiple identical *Select* commands an interrogator can asymptotically single out all tags matching the selection criteria even though tags may undergo short-term RF fades.
- ◆ A *Query* command uses **inventoried** and **SL** to decide which tags participate in an inventory. Interrogators may inventory and access **SL** or **~SL** tags, or they may choose to ignore the **SL** flag entirely.

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Inventory



- ◆ The inventory command set includes *Query*, *QueryAdjust*, *QueryRep*, *ACK*, and *NAK*.
- ◆ *Query* initiates an inventory round and contains a slot-count parameter Q . Upon receiving a *Query* participating tags shall pick a random value in the range $(0, 2^Q - 1)$, inclusive, and shall load this value into their slot counter. Tags that pick a zero shall transition to the **reply** state and reply immediately. Tags that pick a nonzero value shall transition to the **arbitrate** state and await a *QueryAdjust* or a *QueryRep* command.

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Single Tag Reply inventory sequence



- ◆ The tag backscatters an RN16 as it enters **reply**,
- ◆ The interrogator acknowledges the tag with an *ACK* containing this same RN16,
- ◆ The acknowledged tag transitions to the **acknowledged** state, backscattering its PC (Protocol Control), UII (Unique Item Identifier), and CRC-16,
- ◆ The interrogator issues a *QueryAdjust* or *QueryRep*, causing the identified tag to invert its **inventoried** flag (i.e. $A \rightarrow B$ or $B \rightarrow A$) and transition to **ready**, and potentially causing another tag to initiate a query-response dialog with the interrogator.
- ◆ If the tag fails to receive the *ACK* within a predefined time T_2 , or receives the *ACK* with an erroneous RN16, it shall return to **arbitrate**.

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Multiple Tags



- ◆ If multiple tags reply but the interrogator, by detecting and resolving collisions at the waveform level, can resolve a RN16 from one of the tags, the interrogator can *ACK* the resolved tag. Unresolved tags receive erroneous RN16s and return to **arbitrate** without backscattering their PC, UII, and CRC-16.
- ◆ If the interrogator sends a valid *ACK* (i.e. an *ACK* containing the correct RN16) to the tag in the **acknowledged** state the tag shall re-backscatter its PC, UII, and CRC-16.
- ◆ At any point the interrogator may issue a *NAK*, in response to which all tags in the inventory round shall return to **arbitrate** without changing their **inventoried** flag.

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QueryAdjust or QueryRep



- ◆ After issuing a *Query* to initiate an inventory round, the interrogator typically issues one or more *QueryAdjust* or *QueryRep* commands.
- ◆ *QueryAdjust* repeats a previous *Query* and may increment or decrement Q , but does not introduce new tags into the round.
- ◆ *QueryRep* repeats a previous *Query* without changing any parameters and without introducing new tags into the round.
- ◆ An inventory round can contain multiple *QueryAdjust* or *QueryRep* commands.
- ◆ At some point the interrogator will issue a new *Query*, thereby starting a new inventory round.

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QueryAdjust: 1



- ◆ Tags in the **arbitrate** or **reply** states that receive a *QueryAdjust* first adjust Q (increment, decrement, or leave unchanged), then pick a random value in the range $(0, 2^Q - 1)$, inclusive, and load this value into their slot counter. Tags that pick zero shall transition to the **reply** state and reply immediately. Tags that pick a nonzero value shall transition to the **arbitrate** state and await a *QueryAdjust* or a *QueryRep* command.

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QueryAdjust: 2



- ◆ Tags in the **arbitrate** state decrement their slot counter every time they receive a *QueryRep*, transitioning to the **reply** state and backscattering an RN16 when their slot counter reaches 0000h. Tags whose slot counter reached 0000h, who replied, and who were not acknowledged (including tags that responded to the original *Query* and were not acknowledged) shall return to **arbitrate** with a slot value of 0000h and shall decrement this slot value from 0000h to 7FFFh at the next *QueryRep*, thereby effectively preventing subsequent replies until the tag loads a new random value into its slot counter. Tags shall reply at least once in $2^Q - 1$ *QueryRep* commands.

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Q

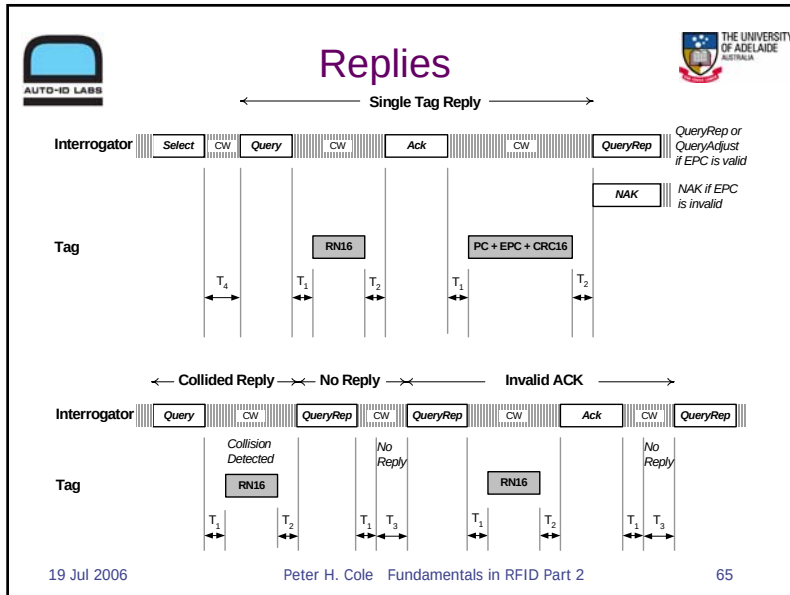


- ◆ Although tag inventory is based on a random protocol, the Q -parameter affords network control by allowing an interrogator to regulate the probability of tag responses. Q is an integer in the range $(0, 15)$; thus, the associated tag-response probabilities range from $2^0 = 1$ to $2^{-15} = 0.000031$.

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Tag Memory: 1

- Tag memory is logically separated into four distinct banks, each of which may comprise zero or more memory words.
- The memory banks are:
 - Reserved memory shall contain the kill and access passwords.
 - UII memory contains a CRC-16, Protocol-Control (PC) bits and a code (such as an UII) that identifies the object to which the tag is or will be attached. Included is a Numbering System Identifier (NSI).

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Tag Memory: 2

- TID memory contains an 8-bit ISO/IEC 15963 allocation class identifier with sufficient identifying information for an interrogator to uniquely identify the custom commands and/or optional features that a tag supports. This identifying information includes a 12-bit tag mask-designer identifier, and a 12-bit tag model number. Tags may also contain tag- and vendor-specific data (for example, a tag serial number).
- User memory allows user-specific data storage. The memory organization is user-defined.

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Access and Kill Passwords

- Kill password
 - 32-bit value stored in Reserved memory
 - Default (unprogrammed) value is zero.
 - An interrogator uses a tag's kill password once, to kill the tag and render it silent thereafter.
- Access password
 - 32-bit value stored in Reserved memory.
 - Default (unprogrammed) value is zero.
 - Tags with a nonzero access password shall require an interrogator to issue this password before transitioning to the secured state.

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Protocol Control Bits: 1

- ◆ The PC bits contain physical-layer information that a tag backscatters with its UII during an inventory operation.
- ◆ There are 16 PC bits, stored in UII memory
 - Bits 10h – 14h: The length of the (PC + UII) that a tag backscatters, in words:
 - ◆ 000002: One word (addresses 10h to 1Fh in UII memory).
 - ◆ 000012: Two words (addresses 10h to 2Fh in UII memory).
 - ◆ 000102: Three words (addresses 10h to 3Fh in UII memory).
 - ◆
 - ◆ 111112: 32 words (addresses 10h to 20Fh in UII memory).

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Protocol Control Bits: 2

- Bits 15h – 16h: RFU (shall be set to 002).
- Bits 17h – 1Fh: A numbering system identifier (NSI) whose default value is 0000000002 and which may include an AFI as defined in ISO/IEC 15961. The MSB of the NSI is stored in memory location 17h. If bit 17h contains a logical 0, then PC bits 18h – 1Fh shall contain an EPCglobal™ Header as defined in the EPC™ Tag Data Standards. If bit 17h contains a logical 1, then PC bits 18h – 1Fh shall contain the entire AFI defined in ISO/IEC 15961.
- ◆ The default (unprogrammed) PC value shall be 0000h.

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Sessions: 1

- ◆ Interrogators support and tags provide 4 sessions (denoted S0, S1, S2, and S3).
- ◆ Tags participate in one and only one session during an inventory round.
- ◆ Two or more interrogators can use sessions to independently inventory a common tag population
- ◆ Tags maintain an independent **inventoried** flag for each session.
 - Each of the four **inventoried** flags has two values, denoted *A* and *B*.
 - At the beginning of each and every inventory round an interrogator chooses to inventory either *A* or *B* tags in one of the four sessions.

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Sessions: 2

- Tags participating in an inventory round in one session shall neither use nor modify the **inventoried** flag for a different session.
- The **inventoried** flags are the only resource a tag provides separately and independently to a given session; all other tag resources are shared among sessions.
- ◆ Sessions allow tags to associate a separate and independent **inventoried** flag to each of several readers.
- ◆ After singulating a tag an interrogator may issue a command that causes the tag to invert its **inventoried** flag for that session (i.e. *A*→*B* or *B*→*A*).

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Example, 2 interrogators



- Interrogator #1 powers-on, then
 - It initiates an inventory round during which it singulates *A* tags in session S2 to *B*,
 - It powers off.
- Interrogator #2 powers-on, then
 - It initiates an inventory round during which it singulates *B* tags in session S3 to *A*,
 - It powers off.
- This process repeats until interrogator #1 has placed all tags in session S2 into *B*, after which it inventories the tags in session S2 from *B* back to *A*.
- Similarly, interrogator #2 places all tags in session S3 into *A*, after which it inventories the tags in session S3 from *A* back to *B*.
- By this multi-step procedure each interrogator can independently inventory all tags in its field, regardless of the initial state of their **inventoried** flags

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Tag Selected Flag: 1



- Tags implement a selected flag, **SL**, which an interrogator may assert or deassert using a *Select* command.
- The Sel parameter in the *Query* command allows an interrogator to inventory tags that have **SL** either asserted or deasserted (i.e. **SL** or **~SL**), or to ignore the flag and inventory tags regardless of their **SL** value. **SL** is not associated with any particular session; **SL** applies to all tags regardless of session.

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Tag Selected Flag: 1



- A tag's **SL** flag has a persistence time.
 - A tag will power-up with its **SL** flag either asserted or deasserted, depending on the stored value, unless the tag has lost power for a time greater than the **SL** persistence time, in which case the tag shall power-up with its **SL** flag deasserted (set to **~SL**).
 - A tag is capable of asserting or deasserting its **SL** flag in 2 ms or less, regardless of the initial flag value. A tag refreshes its **SL** flag when powered, meaning that every time a tag loses power its **SL** flag will have a persistence time

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Commands



- *QueryRep*
- *ACK*
- *Query*
- *QueryAdjust*
- *Select*
- *NAK*
- *Req_RN*
- *Read*
- *Write*
- *Kill*
- *Lock*
- *Access*
- *BlockWrite*
- *BlockErase*

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Dense Reader Operation



◆ Three different methods that use frequency plans or time-division multiplexing (TDM) to minimize or eliminate interrogator-on-tag collisions.

- **Method 1: Single-channel regulatory environment**
- **Method 2: Multi-channel regulatory environment with adjacent-channel backscatter**
- **Method 3: Multi-channel regulatory environment with channel-boundary backscatter**

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Single-channel regulatory environment



- ◆ Interrogator transmissions and tag responses shall be separated temporally
- Synchronized interrogators first commanding tags,
 - then all interrogators transmitting CW and listening for tag responses.
 - Interrogator signalling shall use DSB-ASK modulation with $T_{\text{ari}}=25\mu\text{s}$, or PR-ASK or SSB-ASK modulation with $T_{\text{ari}}=12.5\mu\text{s}$.
 - Tag backscatter may be 20 kbps on an 80 kHz subcarrier ($M=4$); alternatively, tag backscatter may be selected from a variety of modes and data rates.

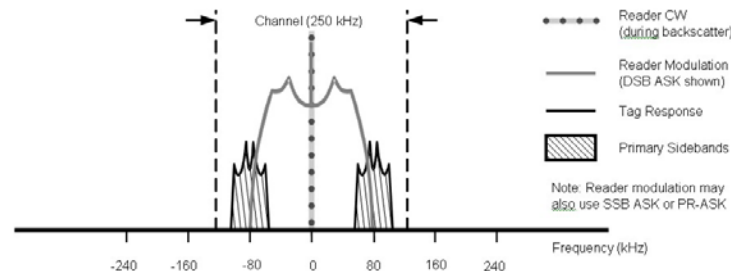
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A: Subcarrier Spectral Allocation (CEPT: Single Channel)



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Multi-channel regulatory environment with adjacent-channel backscatter



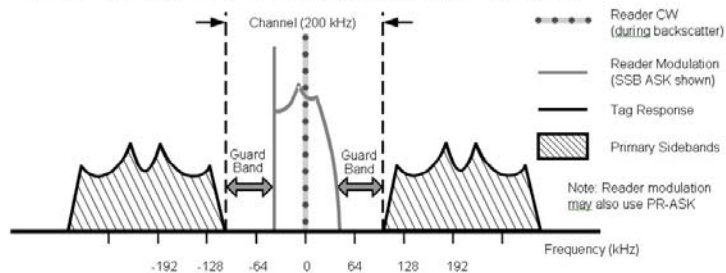
- ◆ Interrogator transmissions and tag responses shall be separated spectrally
- Interrogator transmissions located in even-numbered channels
 - Tag backscatter located in odd-numbered channels.
 - Interrogator signalling shall use PR-ASK or SSB-ASK modulation with $T_{\text{ari}}=25\mu\text{s}$
 - Tag backscatter shall be 53.3 or 26.7 kbps data on a 213.3 kHz subcarrier ($M=4$ or $M=8$, respectively).

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B: Subcarrier Spectral Allocation (CEPT: Multiple Channels)



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Multi-channel regulatory environment with channel-boundary backscatter

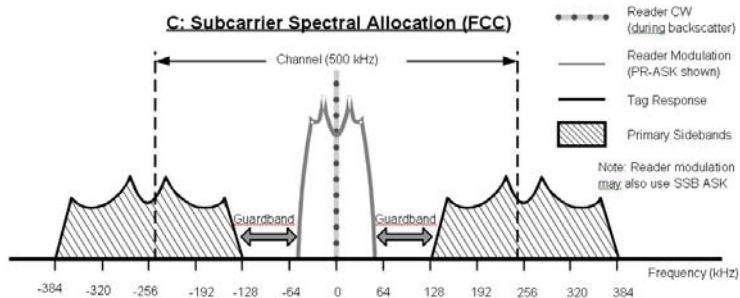
- Interrogator transmissions and tag responses shall be separated spectrally, with interrogator transmissions centred in channels and tag backscatter situated at channel boundaries.
- Interrogator signalling shall use PR-ASK or SSB-ASK modulation with $T_{\text{ari}}=25\mu\text{s}$
- Tag backscatter shall be 64 or 32 kbps data on a 256 kHz subcarrier ($M=4$ or $M=8$, respectively).

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C: Subcarrier Spectral Allocation (FCC)



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PART 11

DENSE READER PROBLEMS AND ISSUES

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Associated document



See also text document “Dense Reader Problems in RFID deployment”

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Dense reader issues



- ◆ High reader and tag density
- ◆ Clarification of terms desirable

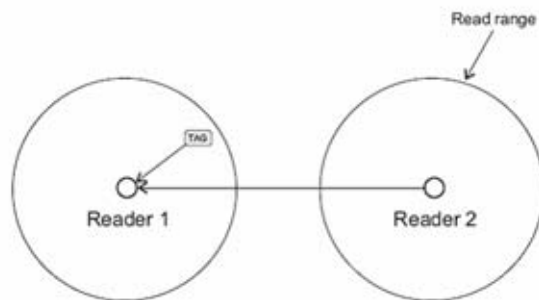
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Reader interference with tag reply



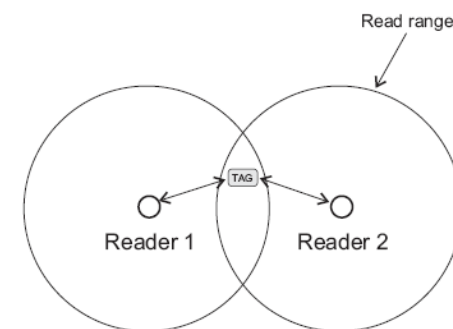
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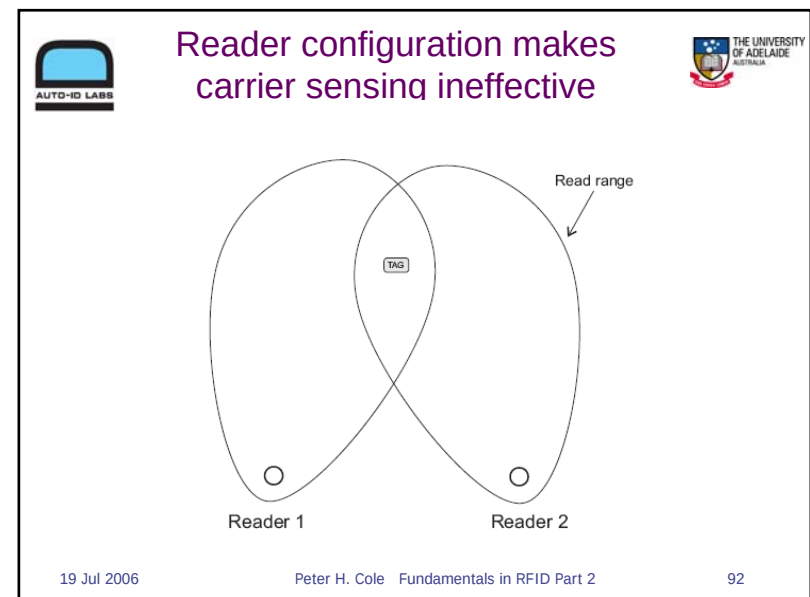
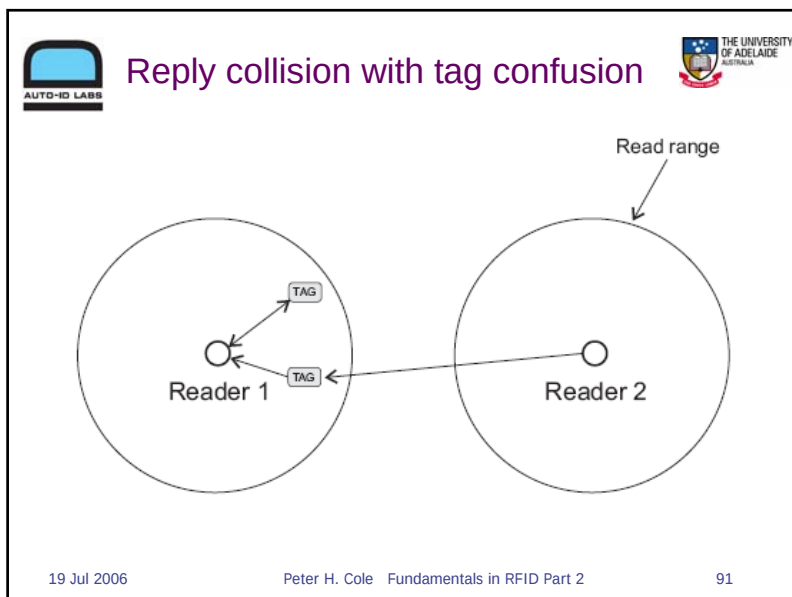
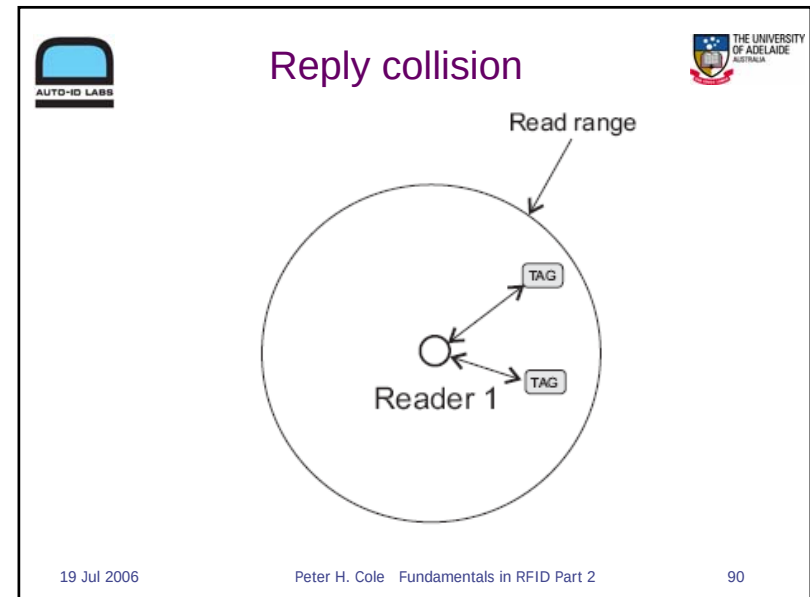
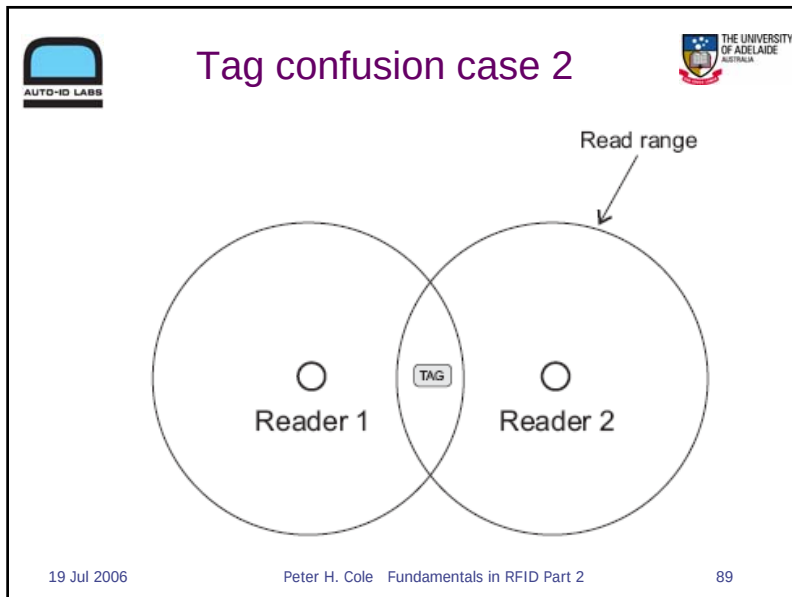
Tag confusion case 1



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Significant issues



- ◆ Separation of transmit and receive channels
- ◆ Organisation of interrogator channels
- ◆ Listen before talk regulations
- ◆ Synchronisation the solution

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The ETSI LBT regulation



Prior to Transmission, the interrogator must listen for the presence of another signal within its intended sub-band of transmission. The listen time shall comprise a fixed period of 5 ms plus a random time of 0 ms to 5 ms in 11 steps. If the sub-band is free the random time shall be set to 0 ms.

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Solution and improvements



- ◆ Reader synchronisation
- ◆ Spatial and frequency allocation of readers
- ◆ Reduction of output power
- ◆ Reduction of overall talking time
- ◆ Absorbing materials
- ◆ Sensors to turn off unwanted readers

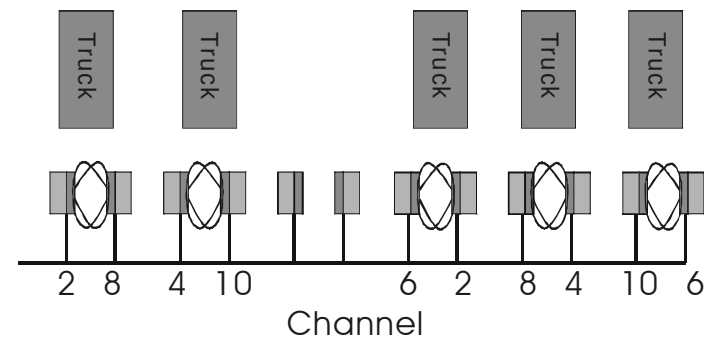
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Illustration



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PART 12

EPCglobal NETWORK ELEMENTS AND FUNCTIONS

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Transformation to Laboratories and EPCglobal



◆ Transformed...

- ...26 October 2003

◆ Auto-ID Labs

- Performs fundamental research related to EPC System
- Builds communities not already using EPC System

◆ EPC Global

- Manages and develops standards
- Markets EPC System

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EPCglobal network: outline



◆ Electronic product code

- Formats for various applications

◆ ID system

- Tags and readers

◆ EPC middleware

- Replaces the savant
- ALE engine and interfaces
- Performs filtering a data routing for clients

◆ Discovery services

- ONS discovery service
- Discovery services for events

◆ EPC information services

- Enables users to securely exchange information with trading partners

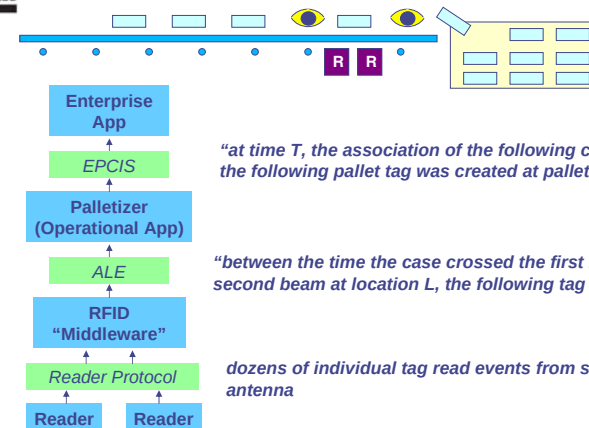
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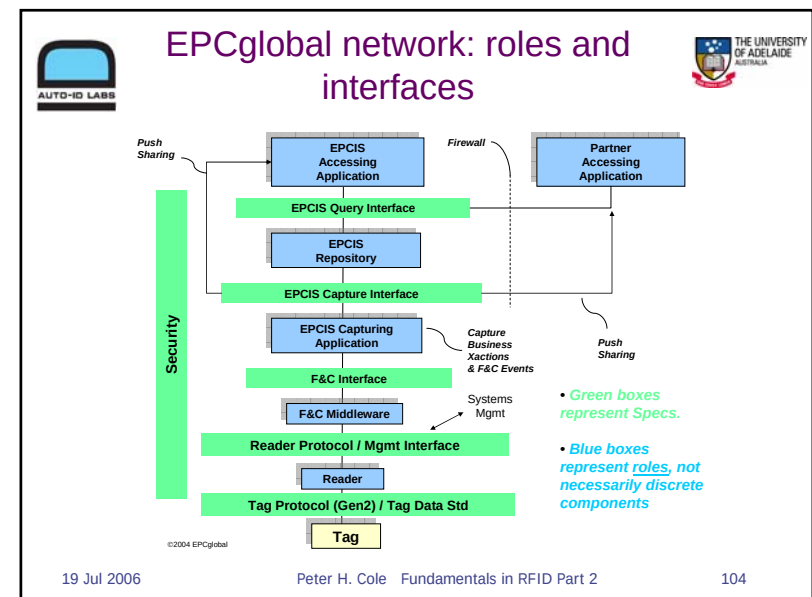
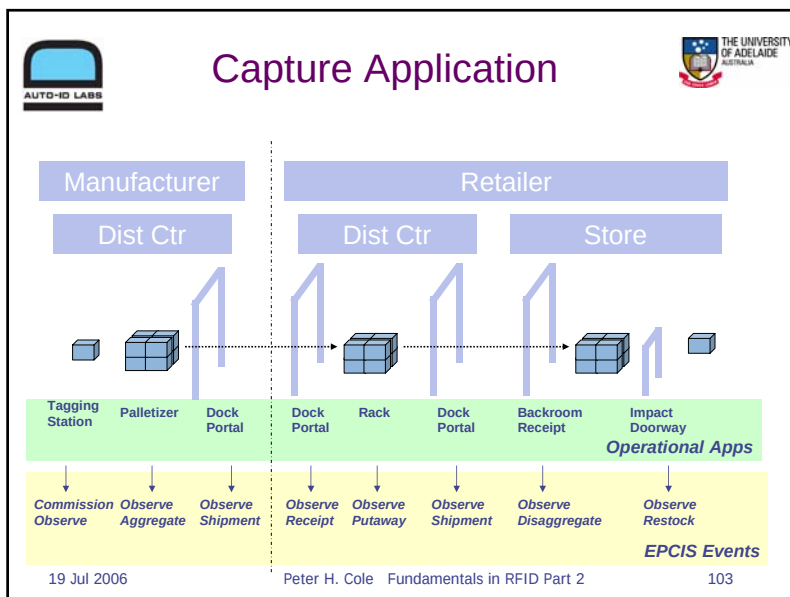
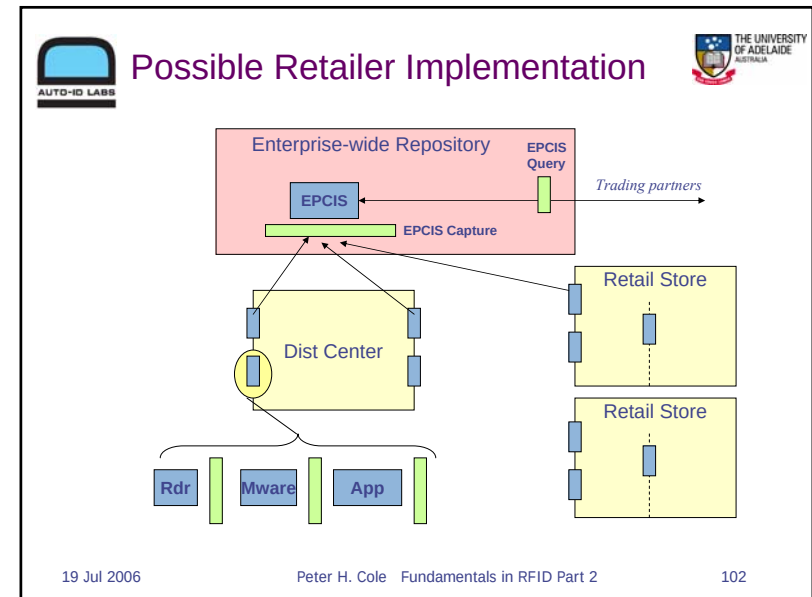
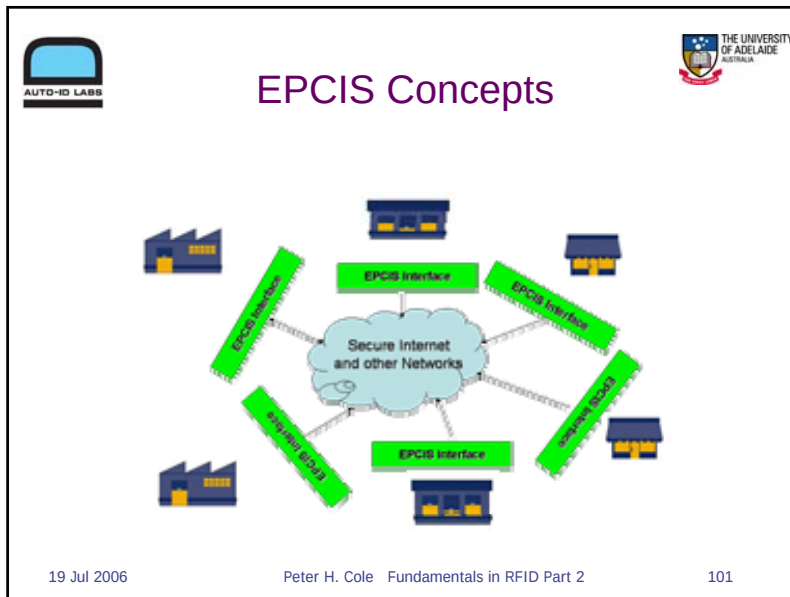
EPC Event Layers



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PART 13

SECURITY AND AUTHENTICATION

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Associated documents



- ◆ A Security and Authentication Primer
- ◆ Definition of Terms used by the Adelaide Auto-ID Lab in Anti-Counterfeiting Studies

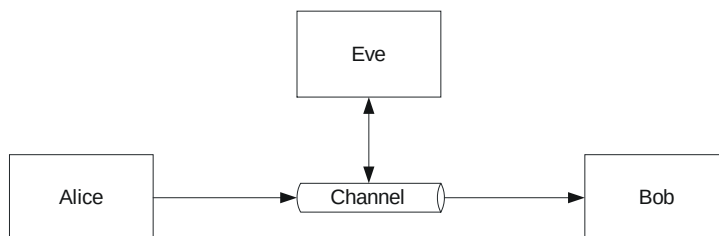
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Encryption model



Alice and Bob trust each other. The channel is assumed insecure. Can Alice be sure only Bob reads her messages? Can Bob be sure a message was sent by Alice and it was not altered? Can either party deny they sent or received a message? Thus we want to provide security services as defined on the next slide.

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Security services



- ◆ **Confidentiality** Only authorised parties receive information
- ◆ **Authentication** The ability of a party to be sure the message is from a claimed source
- ◆ **Integrity** Assures us a message is not altered on the way
- ◆ **Non-repudiation** Proof of transmission and reception
- ◆ **Access Control** Restricts and controls access to a system
- ◆ **Availability** Provides means to assure a system is available when needed

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Attacks



- ◆ Ciphertext-only attack
- ◆ Known-plaintext attack
- ◆ Chosen-plaintext attack
- ◆ Adaptive chosen-plaintext attack
- ◆ Chosen-ciphertext attack
- ◆ Adaptive chosen-ciphertext attack
- ◆ Known-key attack
- ◆ Replay attack
- ◆ Impersonation attack
- ◆ Dictionary attack

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Security models



Apply to cryptographic primitives

- ◆ **Unconditional security** Perfect security, assumes unrestricted computational power of an adversary
- ◆ **Computational security** No known algorithm to break it within polynomial time
- ◆ **Practical security** No breaking algorithm within N operations, with N chosen to be high. Modern primitives offer practical security.
- ◆ **Provable security** Possible to show the complexity of breaking a primitive is equivalent to solving a well known supposedly hard mathematical problem

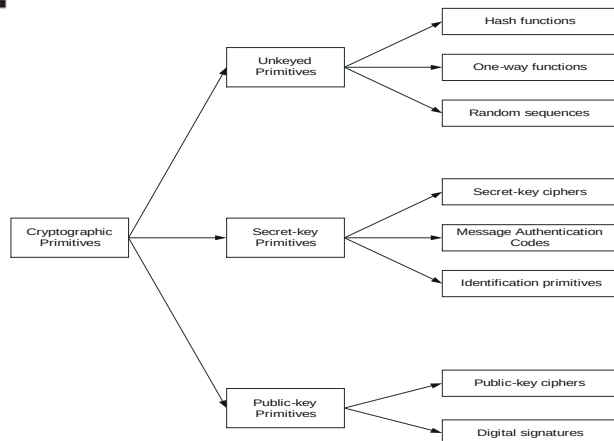
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Cryptographic primitives



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Unkeyed primitives



- ◆ **Hash functions**
 - A hash function is a computationally efficient function mapping binary strings of arbitrary length to binary strings of some fixed length, called hash-values
- ◆ **One way functions** Easy to compute but hard to invert
- ◆ **Trapdoor functions** One way, but with the additional property that given special information it is possible to calculate the inversion
- ◆ **Random number generators** Are an unkeyed primitive in cryptography. They are used for keys in public key cryptography, session keys in secret key cryptography, random sequences (nonces) in cryptographic protocols

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Secret key primitives



Based on keys secretly shared between trusted parties.

- ◆ Secret key ciphers
 - Block ciphers
 - Stream ciphers
- ◆ Message authentication codes
- ◆ Identification primitives
 - Fixed passwords
 - One time passwords
 - Challenge response identification

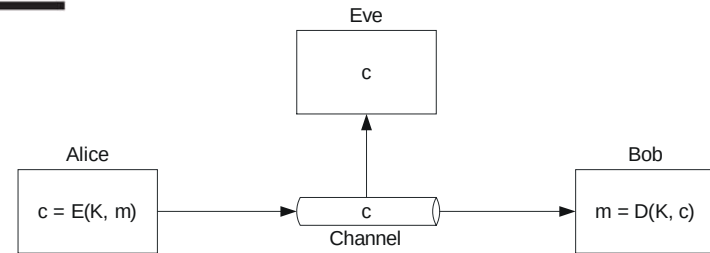
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Secret key encryption



Alice and Bob agree, before the start of transmission, on a secret key pair known only to them. Each key may easily be derived from the other. Often they are the same, denoted by K above. c is the ciphertext, at least as large as the message m . There is the key distribution problem. Block ciphers are a special case. Examples are DES, 3DES, and AES: block size of 128 bits; key sizes of 128, 192 or 256 bits. No known attacks (other than side channel attacks).

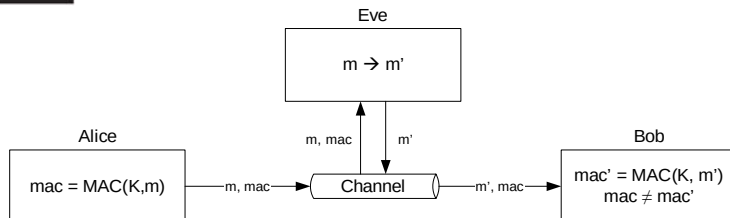
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Message authentication codes



Alice generates a message authentication code mac by applying a MAC function which processes the message and shared secret key K , and transmits both. Bob also generates the mac and compares it with what he received. Eve can change the message but Bob will detect that. Data integrity and data origin authentication are provided.

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Public key cryptography



- ◆ Public key ciphers
- ◆ Digital signatures

These form the second group of keyed cryptographic tools. Based on key pairs instead a single shared key. Only one key need be kept secret. Sometimes called asymmetric key systems. The receiving party issues the public encrypting key and keeps to itself the decrypting key.

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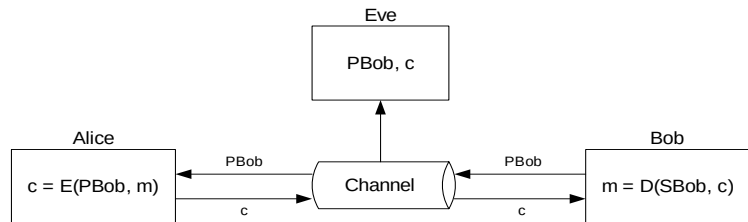
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Public key encryption





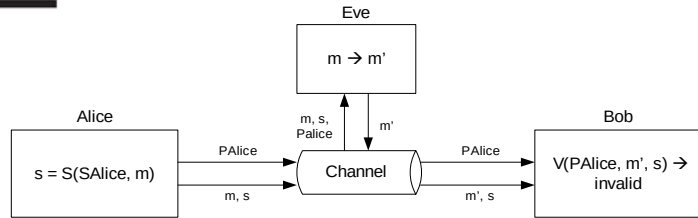
The key pair used in the example is the secret key SBob of Bob and the public key PBob of Bob.

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Digital signatures





Digital signatures provide authentication, authorisation and some non-repudiation services. In contrast with the key usage in public key ciphers, here Alice's private (secret) key is used by her for signing, and her public key is used by Bob for verification. Eve has altered the message m , and Bob can detect that alteration. However, in practice Bobs public key cipher would also be used to encrypt both the message and the signature. In addition, the signature may not be based on the whole message, but on a message digest generated using a hash function.

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Comparison secret and public



- ◆ Secret key
 - More secure for given key size
 - Key distribution problem
 - Multiple correspondent problem
 - Mostly turn out to be symmetric keys
 - ◆ But do not have to be
- ◆ Public key
 - More convenient
 - Longer key sizes needed

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Security and privacy concepts



See page 109 for meanings

- ◆ Security aims
 - 'Confidentiality'
 - 'Integrity'
 - 'Authentication'
 - 'Non-reputation'
 - 'Availability'
- ◆ Privacy aims
 - 'Anonymity'
 - 'Unlinkability'

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Secure RFID challenges



Challenge	Description
Cost	Storage limitation. Silicon area.
Regulations	Radiated power. Frequency of operations. Available bandwidth.
Power Consumption	Power consumption of the label IC circuit. Power required to operate E ² PROM.
Performance	Label performance and system performance goals.
Power disruptions	Sudden loss of power.

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Some new directions



- ◆ **Hash locking** Tag is locked by a random key, but stores a hashed value of this key, and will only respond with this hashed value, which serves as a meta id, and will only be unlocked if it received the original secret key.
- ◆ **Hash chains** Similar but based on two different hash functions H and G and a back end data base. Ensures that tag responses are different for each query.
- ◆ **Re-encryption** In RFID used to denote changing ciphertexts without knowledge of public keys. Complex. See page 139 for details.
- ◆ **Physical uncloneable functions**
- ◆ **Lightweight one time codes**

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PART 14

AUTO-ID RESEARCH

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The Auto-ID Laboratories



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Laboratories research program



- ◆ Seven laboratories around the world
- ◆ Associate laboratories are contemplated
- ◆ Flagship research projects
 - Anti counterfeiting
 - Sensor networks
 - Aerospace and automotive
- ◆ Additional laboratory research projects

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PART 15

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Summary



- ◆ Propagation and coupling theory
- ◆ Antennas for various applications
- ◆ Subthreshold turn on circuits
- ◆ Optimum matching theory
- ◆ Small tags in difficult environments
- ◆ Security and authentication systems
- ◆ High density reader systems
- ◆ Dual band single port antennas
- ◆ Zero power turn on circuits

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Propagation and Coupling Theory

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The field vectors



A full theory of electrodynamics, including the effects of dielectric and magnetic materials, must be based on the four field vectors:

- Electric field vector $\mathcal{E}$
- Magnetic field vector $\mathcal{H}$
- Electric flux density vector $\mathcal{D}$
- Magnetic flux density vector $\mathcal{B}$

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Material state vectors



$\mathcal{P}$ for a dielectric material

$\mathcal{M}$ for a magnetic material

$$\mathcal{D} = \epsilon_0 \mathcal{E} + \mathcal{P}$$

$$\mathcal{B} = \mu_0 (\mathcal{H} + \mathcal{M})$$

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Laws in differential form



$$\begin{aligned}\nabla \times \mathcal{E} &= -\frac{\partial \mathcal{B}}{\partial t} \\ \nabla \times \mathcal{H} &= \mathcal{J} + \frac{\partial \mathcal{D}}{\partial t} \\ \nabla \cdot \mathcal{D} &= \rho \\ \nabla \cdot \mathcal{B} &= 0\end{aligned}$$



Vortex



Source

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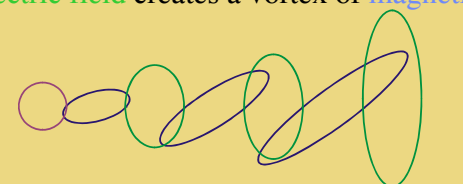
Electromagnetic propagation



Electric current creates a vortex of magnetic field

Magnetic field creates a vortex of electric field

Electric field creates a vortex of magnetic field



Propagation

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Electromagnetic waves



- ◆ They propagate with the velocity of light
 - (Light is an electromagnetic wave)
- ◆ Velocity c is 300,000,000 m/s
- ◆ Wavelength - frequency relation is $c = f\lambda$
- ◆ But not all electromagnetic fields are propagating waves; some are just local energy storage fields
- ◆ This makes a profound difference in how to calculate coupling

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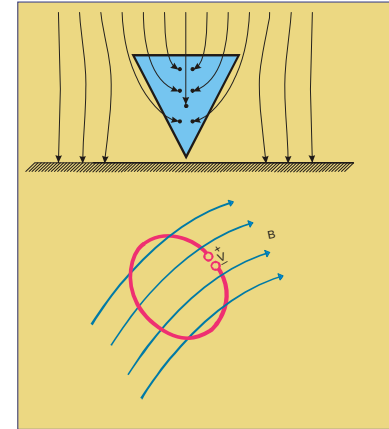
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The basic laws: how they work



- ◆ Gauss's law
 - Electric flux deposits charge
 - Electric field cannot just go past a conductor, it must turn and meet it at right angles
- ◆ Faraday's law
 - Oscillating magnetic flux induces voltage in a loop that it links



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The radian sphere



At $\beta r = 1$, $r = \lambda/2\pi$, and

- The phase factor $e^{-j\beta r}$ is one radian
- Inside this sphere the near field predominates
- Outside this sphere the far field predominates

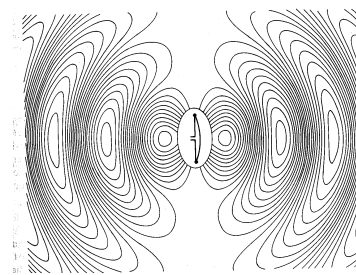
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Near and far field distributions



Electric field launched by an electric dipole
There is also a magnetic field not shown
Note the differences between near and far fields

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Fields of a Magnetic Dipole



$$H_r = \frac{j\beta^3 M}{4\pi} \left(\frac{2}{(\beta r)^2} - \frac{2j}{(\beta r)^3} \right) e^{-j\beta r} \cos \theta$$

$$H_\theta = \frac{j\beta^3 M}{4\pi} \left(\frac{j}{(\beta r)} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3} \right) e^{-j\beta r} \sin \theta$$

$$E_\phi = \frac{j\beta^3 M}{4\pi} \left(\frac{j}{(\beta r)} + \frac{1}{(\beta r)^2} \right) e^{-j\beta r} \sin \theta$$

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Near and far fields



- The far field is an energy propagating field
- Appropriate measure of strength is $0.5 \eta H^2$ (power flowing per unit area)
- The near field is an energy storage field
- Appropriate measure of strength is reactive power per unit volume $0.5 \omega \mu_0 H^2$
- Near field - far field boundary is $\lambda/2\pi$
- Examples 100 kHz 500m; 10 MHz 5m; 1000 MHz 50mm

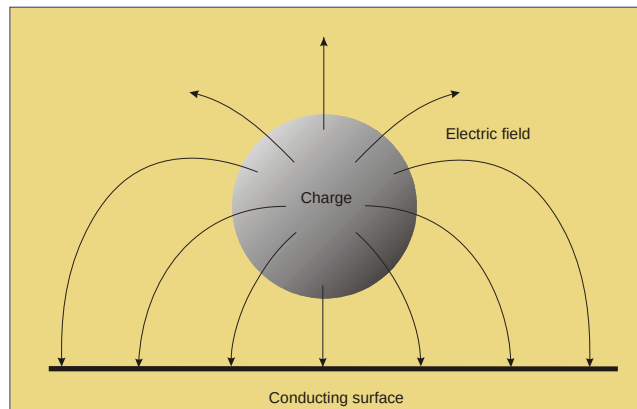
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Boundary Condition: electric field



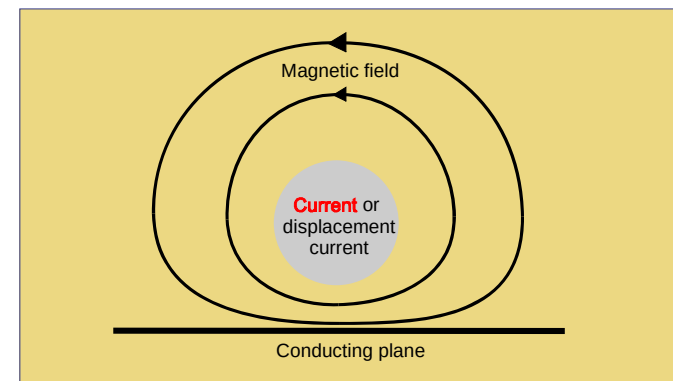
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Boundary Condition: magnetic field



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Near field coupling theory: 1



$$V_c = \frac{\left[\begin{array}{c} \text{Reactive power flowing in the untuned label coil} \\ \text{when it is short circuited} \end{array} \right]}{\left[\begin{array}{c} \text{Volume density of reactive power created by} \\ \text{the interrogator at the label position} \end{array} \right]}$$

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Near field coupling theory: 2



$$V_d = \frac{\left[\begin{array}{c} \text{Reactive power flowing in the inductor} \\ \text{of the interrogator field creation coil} \end{array} \right]}{\left[\begin{array}{c} \text{Volume density of reactive power created by} \\ \text{the interrogator at the label position} \end{array} \right]}$$

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Near field coupling theory: 3



$$\frac{P_2}{P_1} = \frac{V_c}{V_d} Q_1 Q_2$$

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Measures of exciting field



In the far field

$$W_v = \beta S_r$$

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Significant conclusions



- ◆ Coupling volumes for well shaped planar electric and magnetic field labels are size dependent and similar

$$\text{Magnetic } V_c = \frac{L^3}{2} \quad \text{Electric } V_c = \frac{2L^3}{3}$$

- ◆ Radiation quality factors for both types of label formed within a square of side L are size dependent and similar

$$\text{Magnetic } Q_r = \frac{40}{(\beta L)^3} \quad \text{Electric } Q_r = \frac{13}{(\beta L)^3}$$

- ◆ These are calculated results for sensibly shaped antennas

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Optimum operating frequency



The optimum frequency for operation of an RFID system in the far field is the lowest frequency for which a reasonable match to the radiation resistance of the label antenna can be achieved, at the allowed size of label, without the label or matching element losses intruding.

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EAS and Data tags

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Interesting questions



- ◆ Merging of EAS and Data tags

- No fruitful approach yet
- Reasons understood

- ◆ Some new approaches to EAS under investigation

- Micromechanics based

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Sub-threshold turn on circuits

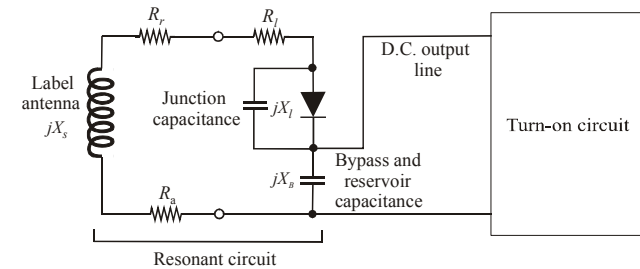
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Transmitter operated systems

- A small voltage or a large is generated from the transmitted power
- A low power consumption circuit, or a zero power consumption circuit, detects that event
- Quality factor and detuning issues arise

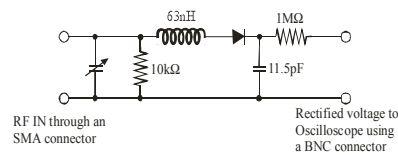


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Experiments on detuning

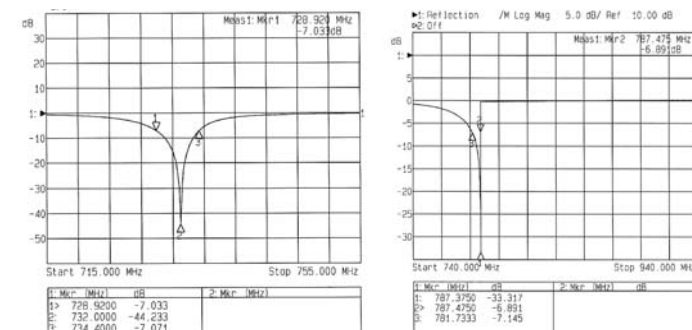


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Low and high power sweeps



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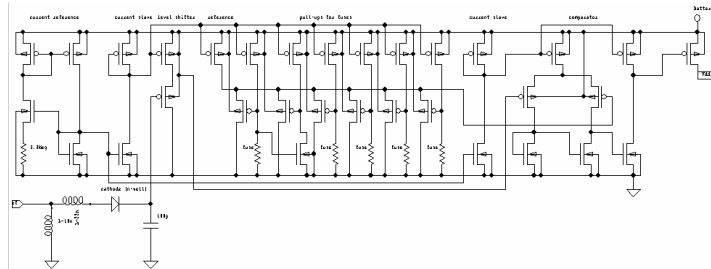
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A low voltage turn on circuit

◆ Sensitivity about 5 mV

◆ Power consumption few nA



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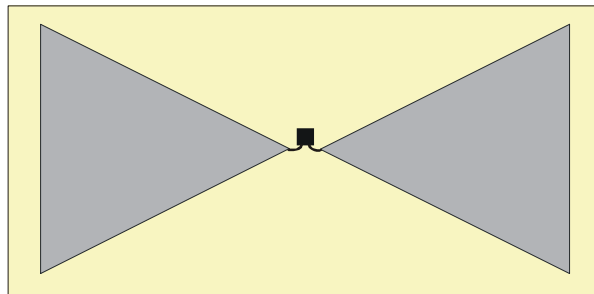
Antennas for various applications

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Electric field bow tie

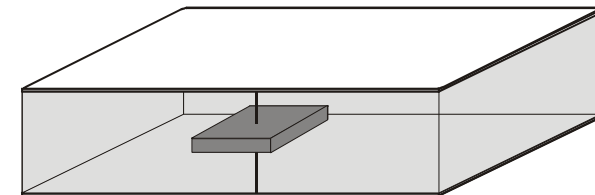


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Electric field box structure



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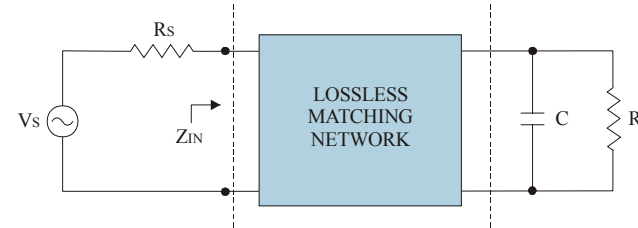
Optimum load matching

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Bode-Fano Limit



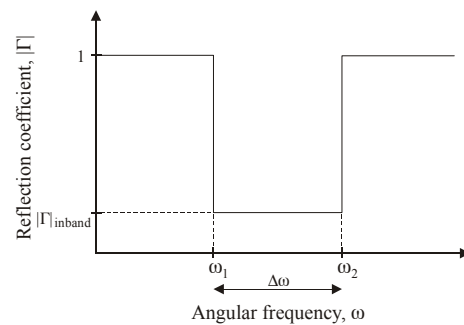
$$\int_0^{\infty} \ln \frac{1}{|\Gamma|} d\omega \leq \frac{\pi}{RC}$$

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Bode-Fano Limit (cont)



$$|\Gamma|_{\text{inband}} \geq e^{-\frac{1}{2\Delta\omega RC}}$$

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Bode-Fano Limit (cont)

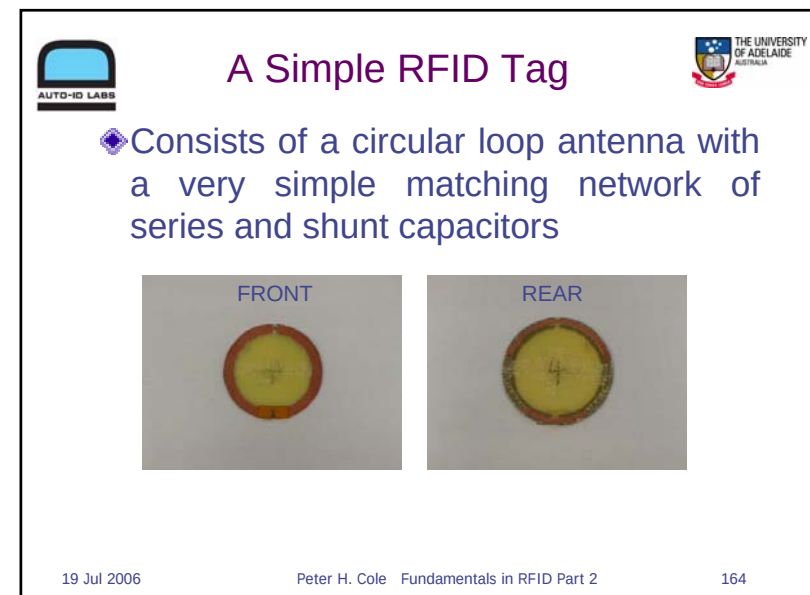
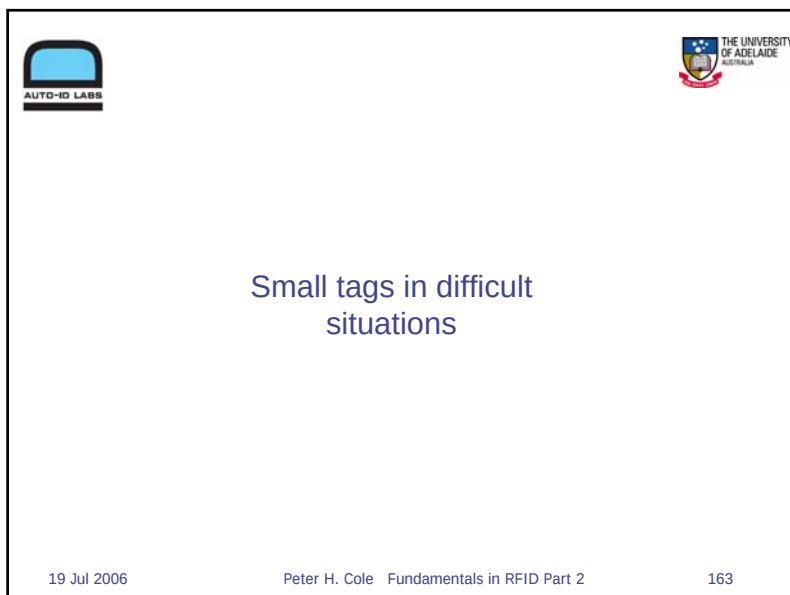
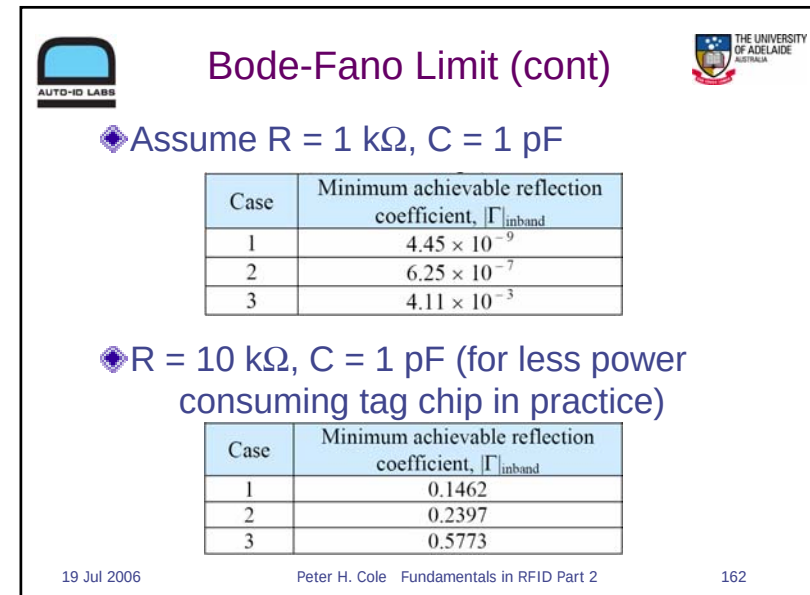
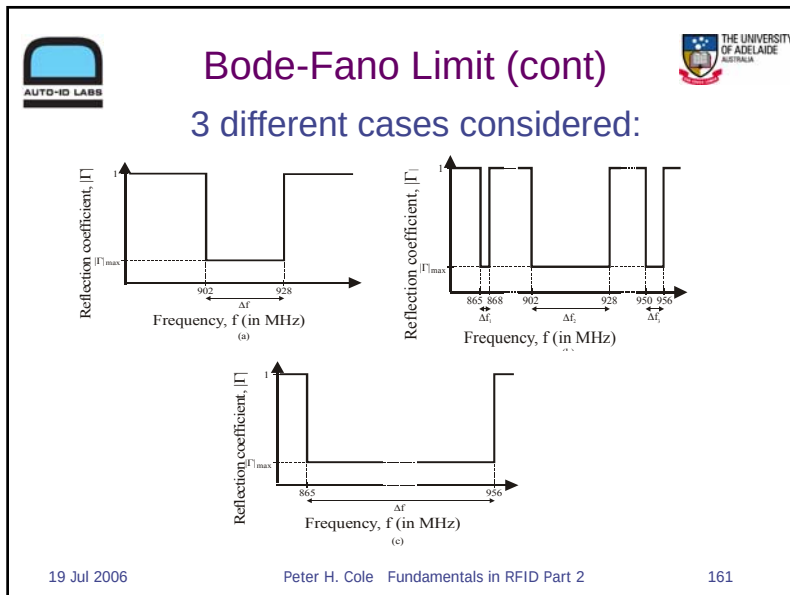
Allocated bandwidths

- ◆ **USA**
 - 902-928 MHz 4W EIRP FHSS
- ◆ **CEPT Countries**
 - 865.6 – 867.6 MHz 2W ERP LBT
- ◆ **South Africa**
 - 865.6 – 867.6 MHz 2W ERP LBT (coming)
 - 917-921 MHz 4W EIRP FHSS
- ◆ **China:**
 - 917 – 922 MHz 2W ERP (Temporary licence can be applied)
- ◆ **Australia:**
 - 920 – 926 MHz 4W EIRP (Temporary licence can be applied)

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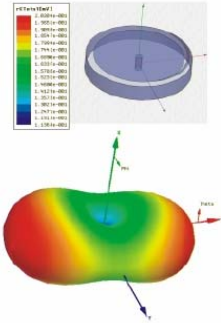
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Tag in metal

Design of an embedded UHF RFID label antenna for a metal structure



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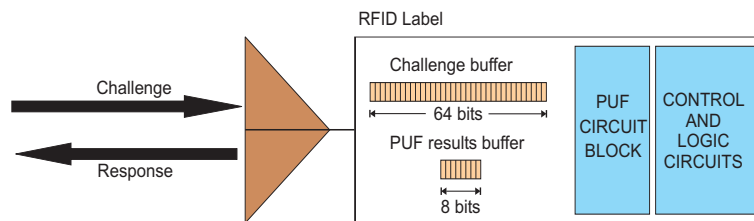
Security and Authentication

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A PUF based Class I/II Label

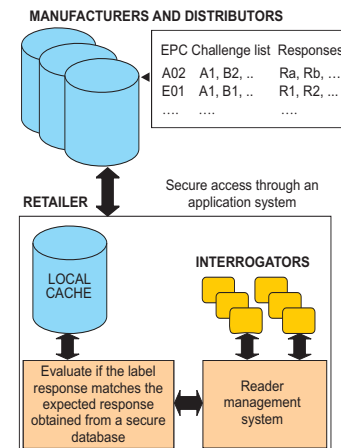


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PUF Based Authentication System



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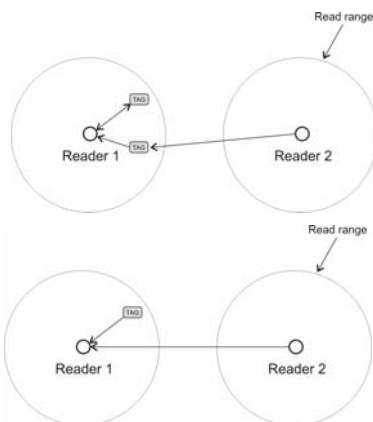
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One time codes

- Lightweight cryptography
- Refreshment under observation

High Density Reader Systems

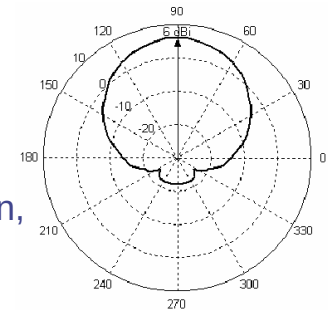
- ◆ Tag confusion
(interference in tag receiver)



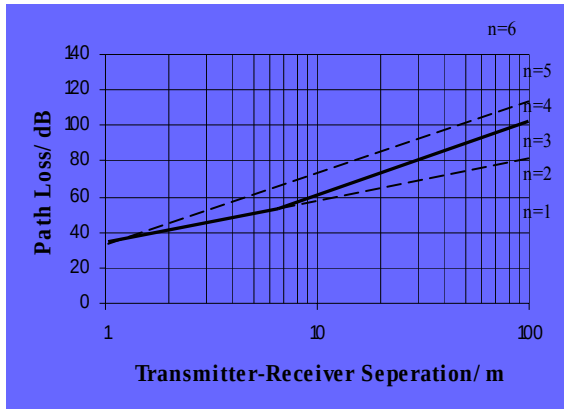
Reader to reader interference

◆ Negligible reflection,
empirically justified.

- ◆ All Antennas deployed have similar gain pattern, orientations may vary



Simulation Model: 1



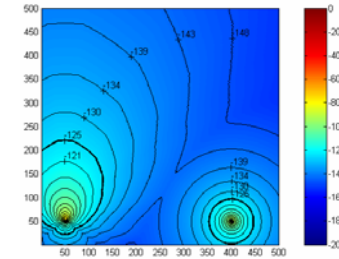
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Frequency Channelling

- Both antennas operating 3 channels away from channel of interest.
- The black line indicates the boundary where a antenna can be positioned to operate in the channel of interest.



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Dual Band Single Port Antennas

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Context and Objectives

- Some applications need benefits of far field
 - Long range
- Some applications need benefits of near field
 - Good spatial discrimination
- Objective is to design antennas that will work equally well at HF (13.56 MHz) and UHF (900 MHz)
- Investigations of whether developments in protocols are needed.

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Results and directions



◆ Working model

- Good antenna
- Good matching to expected loads

◆ Compacting the layout

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Zero power turn on circuits

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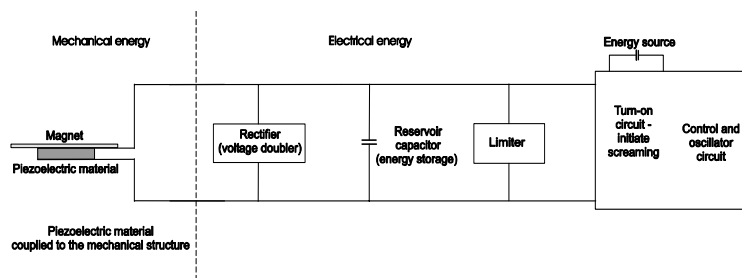
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Zero power turn on concept



- ◆ Low frequency magnetic field vibrates a magnet
- ◆ Piezoelectric converter generates about a volt



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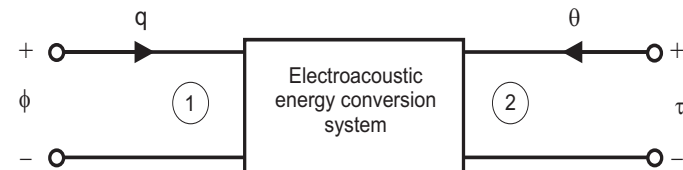
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Electro-acoustic conversion



- ◆ Variables are
 - Torque and angular displacement,
 - charge and voltage
- ◆ Electro-acoustic parameters for substances known; parameters for structures are calculated from the material parameters



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Electroacoustic conversion



◆ Structural parameters appear below

$$\begin{bmatrix} q \\ \theta \end{bmatrix} = \begin{bmatrix} C_{11P} & C_{12P} \\ C_{21P} & C_{22P} \end{bmatrix} \begin{bmatrix} \phi \\ \tau \end{bmatrix}$$

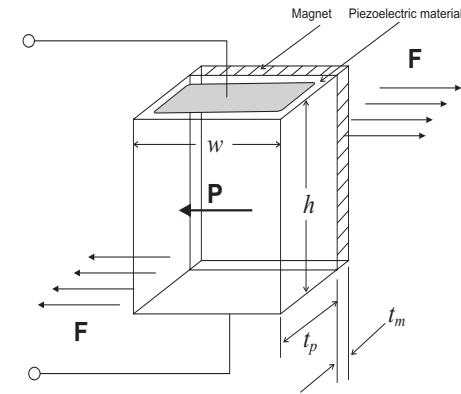
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Structure analysed



Better geometries can be found, but this one does illustrate useful general principles

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Result



◆ Turn-on voltage depends on

- Driving magnetic field
- Electro-acoustic parameters
- Some resonance quality factors

$$V_{TO} = \sqrt{k_{eff}^2 Q_m^2 (Mv\mu_0)^2 |H|^2 C_{22S}^2 \frac{1}{C_J C_{22eff}}}$$

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Definitions of variables



- ◆ V_{TO} r.m.s. voltage generated across the external capacitive load C_J
- ◆ k_{eff} effective electromechanical coupling factor for the piezoelectric structure
- ◆ Q_m mechanical resonance quality factor for the combined magnetic and piezoelectric structure
- ◆ M magnetisation of the magnetic material
- ◆ v volume of the magnetic material
- ◆ H r.m.s value of the oscillating magnetic field
- ◆ C_{22S} total compliance of the structure at the mechanical port
- ◆ C_{22eff} effective compliance of the piezoelectric material at its mechanical port

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Summary



- ◆ Varied research program
- ◆ Partnerships welcome

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PART 15 CONCLUSIONS

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What to take away: 1



- ◆ Simplicity of passive RFID for identity
- ◆ The weakness of the label reply
- ◆ Ubiquity of objects in supply chain
- ◆ Vision of the Auto-ID Center
- ◆ Electric and magnetic field concept
- ◆ Source and vortex concepts
- ◆ Frequency wave length relation $c = f\lambda$

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What to take away: 2



- ◆ Near and far field concepts
- ◆ Radian sphere: size and significance
- ◆ Boundary conditions near meta
- ◆ Behaviour of simple antennas
- ◆ Varieties of fast reading protocol
- ◆ Transformation of Center
- ◆ Auto-ID Labs research

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What to take away: 3



◆ EPCglobal networking concepts

- Standardised EPC
- Standardised readers, tags and protocols
- Standardised communication between roles

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What to take away: 4



◆ The new research targets

- Security and authentication
- Higher functionality tags
- Active tags
- Sensor networks
- Turn on circuits

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Thank you

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